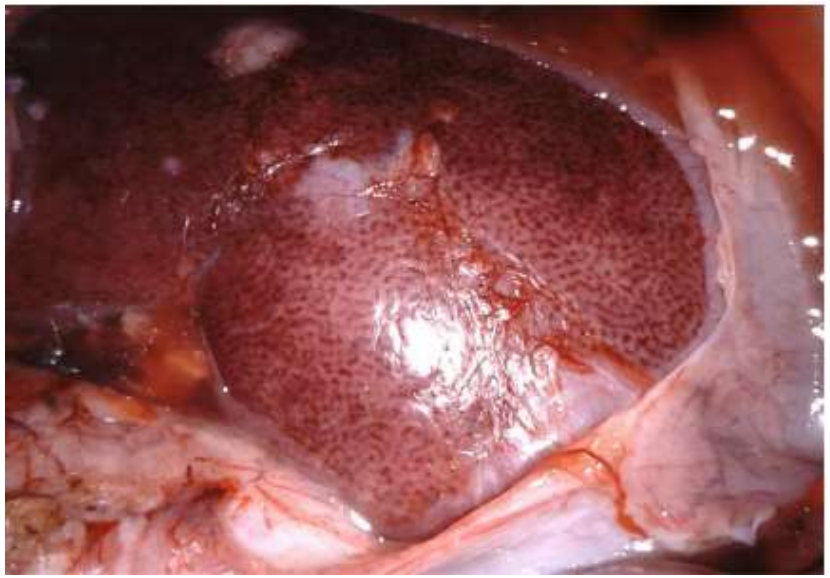
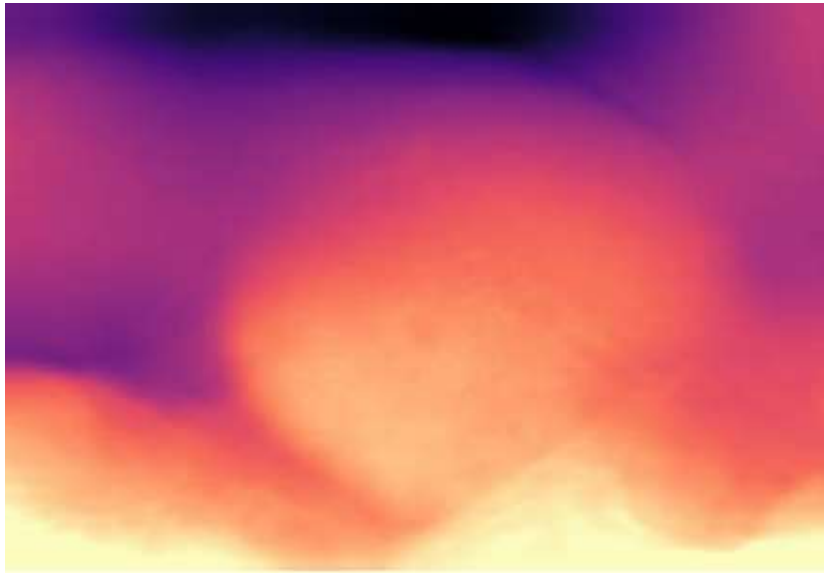
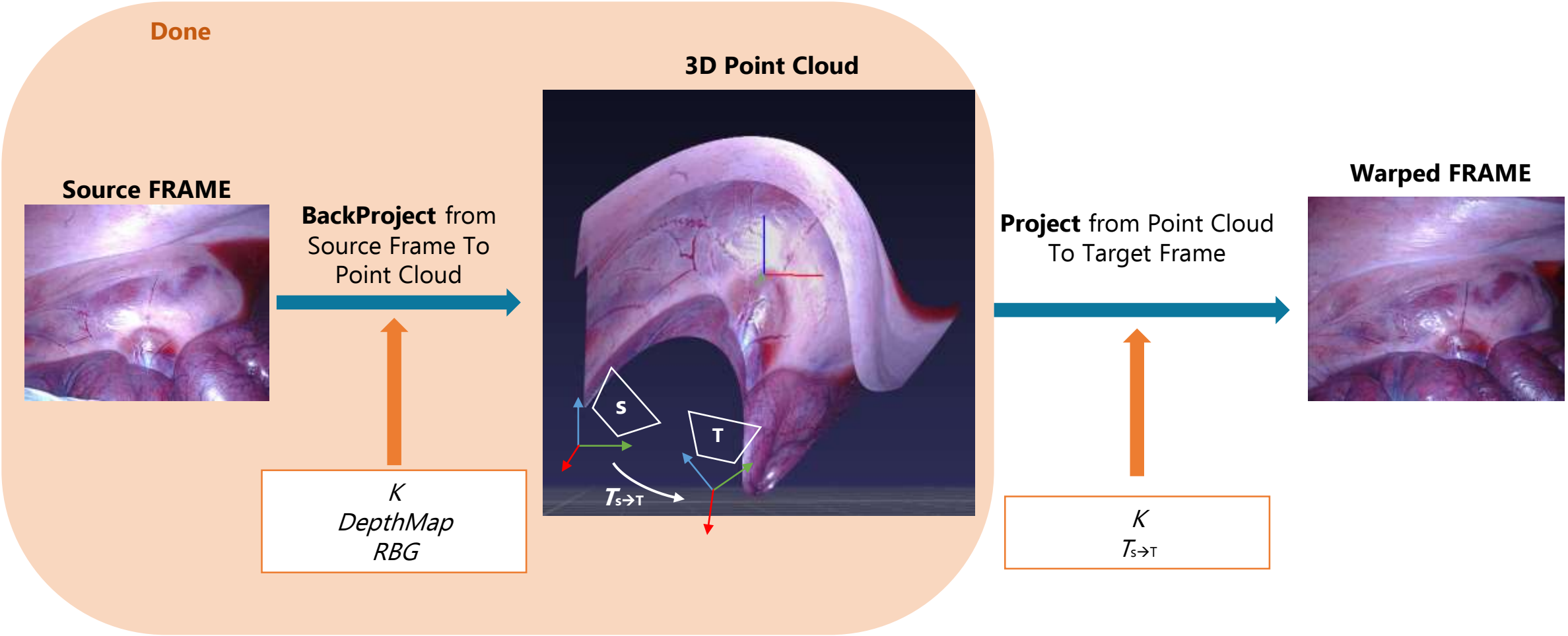
Single-Image Depth Estimation

Projections

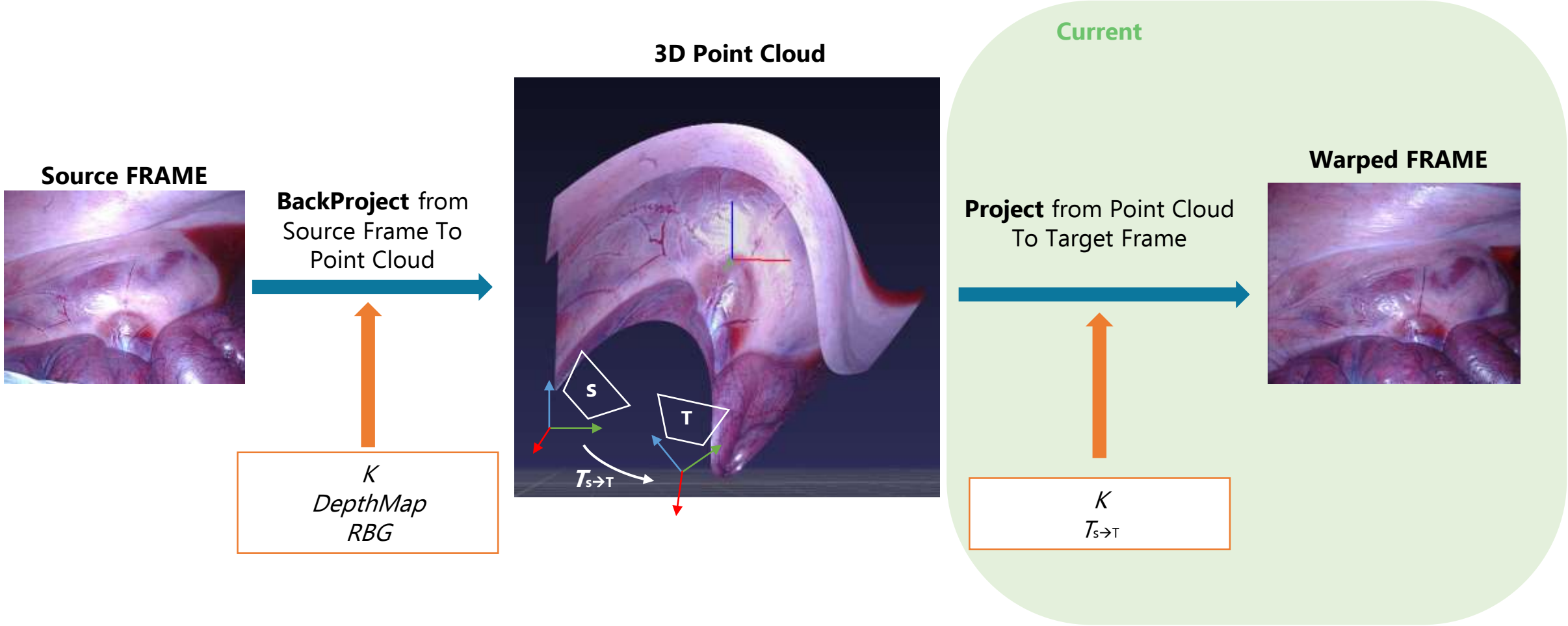




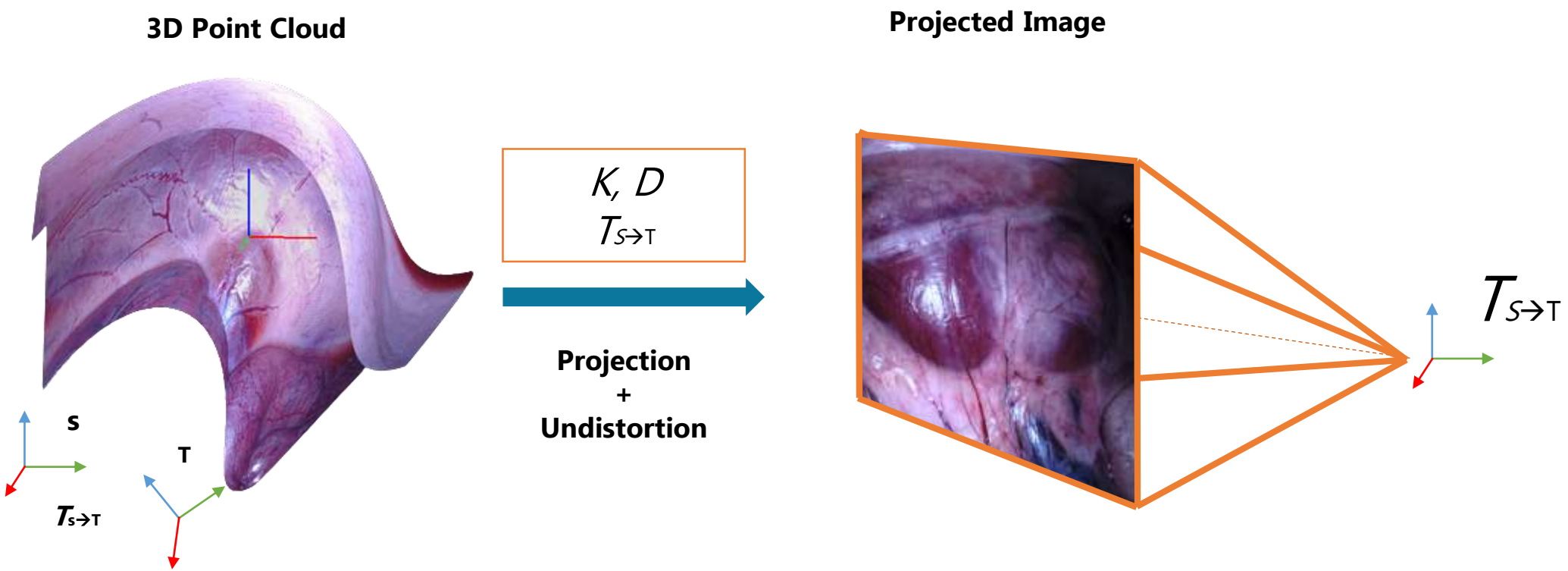
Pipeline


Done
3D Point Cloud
Source FRAME
BackProject from Source Frame To Point Cloud
K
DepthMap
RBG
S
T
$T_{s \to T}$
Project from Point Cloud To Target Frame
K
$T_{s \to T}$
Warped FRAME

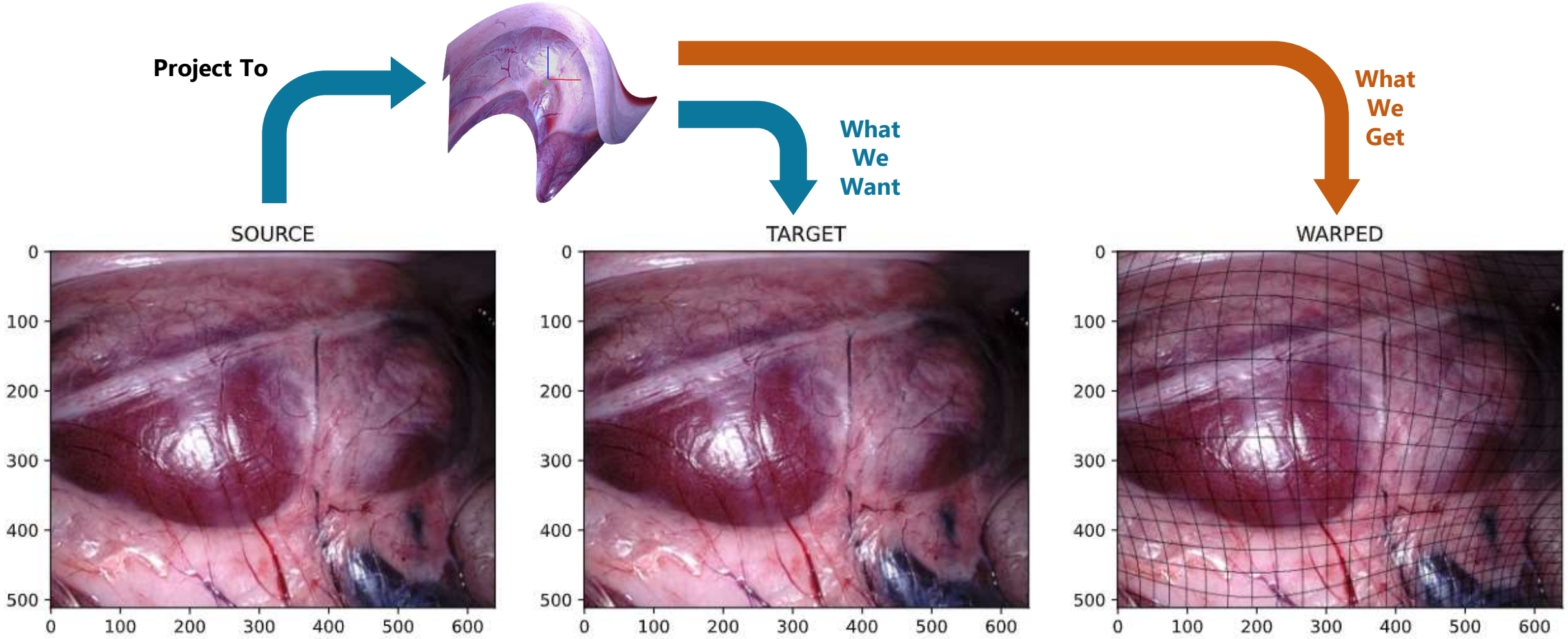
Pipeline


Current
Source FRAME
BackProject from Source Frame To Point Cloud
K
DepthMap
RBG
3D Point Cloud
S
T
$T_{S\rightarrow T}$
Project from Point Cloud To Target Frame
K
$T_{S\rightarrow T}$
Warped FRAME

Projection Framework



Projection Results



Projection Results

- Heavy **Grid Artifacts** → Regularly spaced
  → Following the depthmap contour

Where do they come from? **Rounding errors? Distortions?**

→ *For now, removed manually*

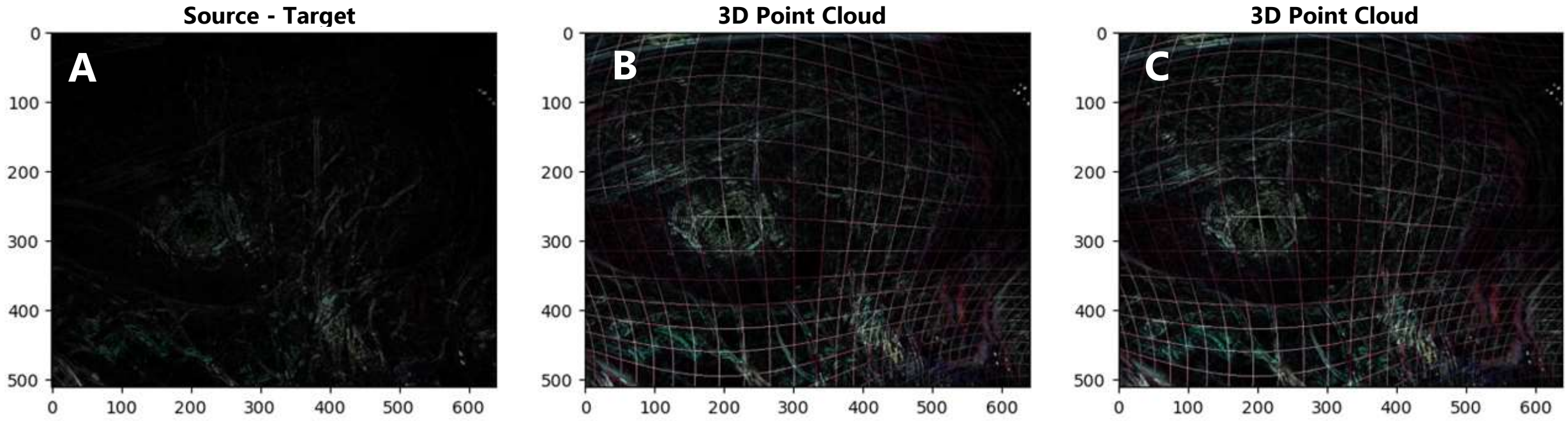
PERFORMANCE: BackProjection + Projection = $\mathbf{0.23 \pm 0.05}\ \boldsymbol{sec}$

→ Might be further reduced, bottleneck is pixel-to-image allocation

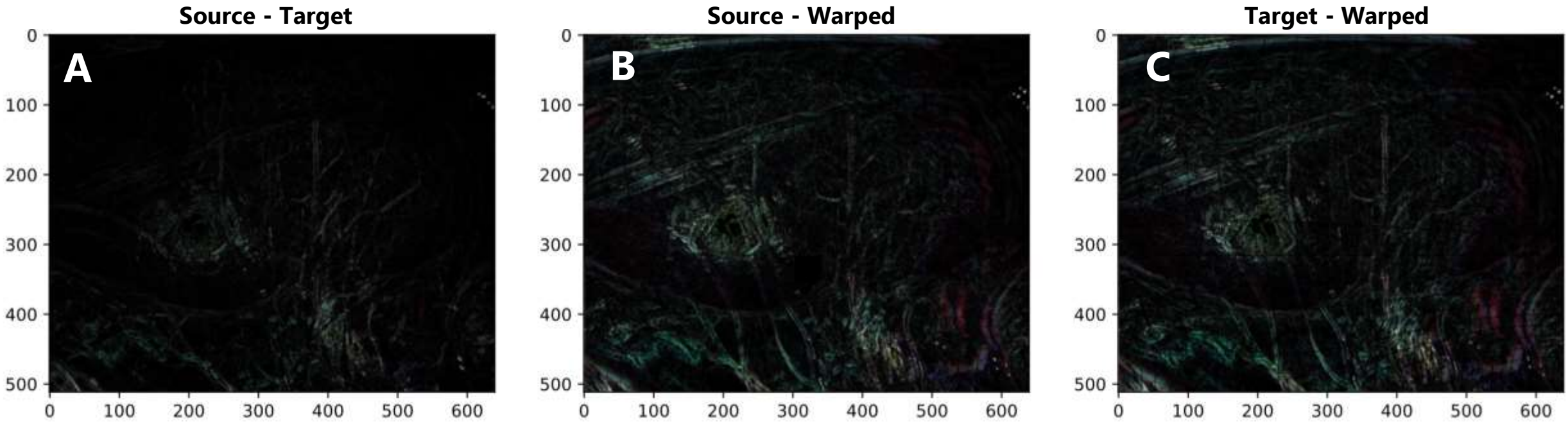
Projection

Frame differences (absolute, not normalized) **with gridline artifacts**

→ *For our purpose,* **C** *should contain information*



Projection

Frame differences (absolute, not normalized) **after gridline artifacts removal**

→ *For our purpose,* **C** *should contain information*